



Evaluating LoRA in Convolutional Neural Networks



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1. Introduction

Training modern Machine Learning (ML) models is computationally and energy expensive, especially under adversarial training, raising sustainability concerns. Neural networks learn input-output mappings through layered weight adjustments, and convolutional neural networks (CNNs) use small kernels to capture local features, excelling at vision tasks. This study uses CNNs to evaluate power consumption and test parameter-efficient methods such as LoRA. We will measure energy and latency systematically, report metrics like Joules per epoch and throughput, and analyze adversarial scenarios such as data poisoning and evasion.

2. Problem & Hypothesis

This research addresses the problem that training CNNs, especially under adversarial training, is time and energy intensive, increasing computational and environmental costs while accuracy and robustness must be maintained. Motivated by the need for scalable and sustainable training, we hypothesize that applying Low-Rank Adaptation (LoRA) to CNNs will effectively help reduce power consumption. By replacing full-rank weight updates with low-rank approximations, we reduce the number of trainable parameters, lower power consumption across epochs, and accelerate convergence with only a marginal trade-off in accuracy. We further hypothesize that incorporating LoRA into adversarial training will mitigate its additional computational overhead while preserving robustness.

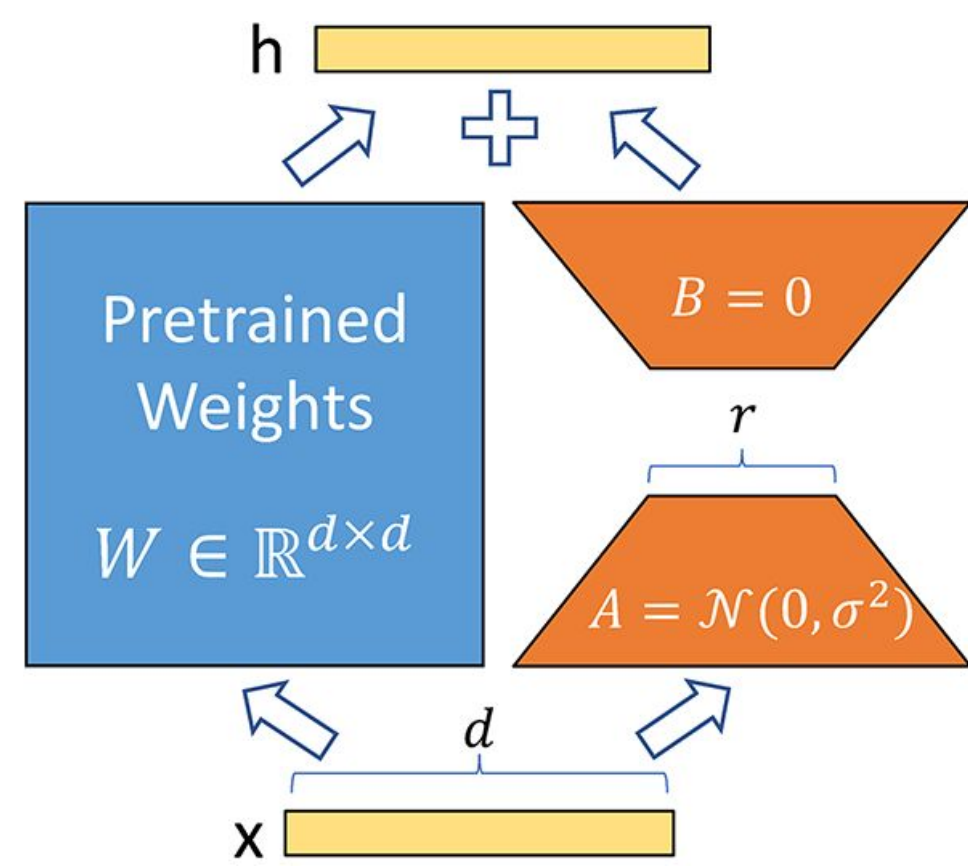


Figure 1. Implementation of LoRA During Training.

3. Objectives

- Quantify the energy-accuracy trade-offs of a LoRA-based model by comparing it against a Full-Parameter Baseline Model, a Fine-tuned Model, and a lightweight baseline on image classification.
- Evaluate the implementation of a set of practical, data-driven guidelines that enable development teams to select optimal training configurations under specific energy or computational budgets.
- Evaluate adversarially trained models with and without LoRA and quantify the trade-offs between energy, accuracy, and robustness.

4. Methodology

Conduct a literature review on LoRa in CNNs, energy tools, and adversarial training to develop strategies for reducing power consumption.

Establish a baseline CNN model and test LoRA-based adversarial models for energy-accuracy trade-offs.

Evaluate defense against data poisoning, evasion, and model inversion while measuring additional energy costs.

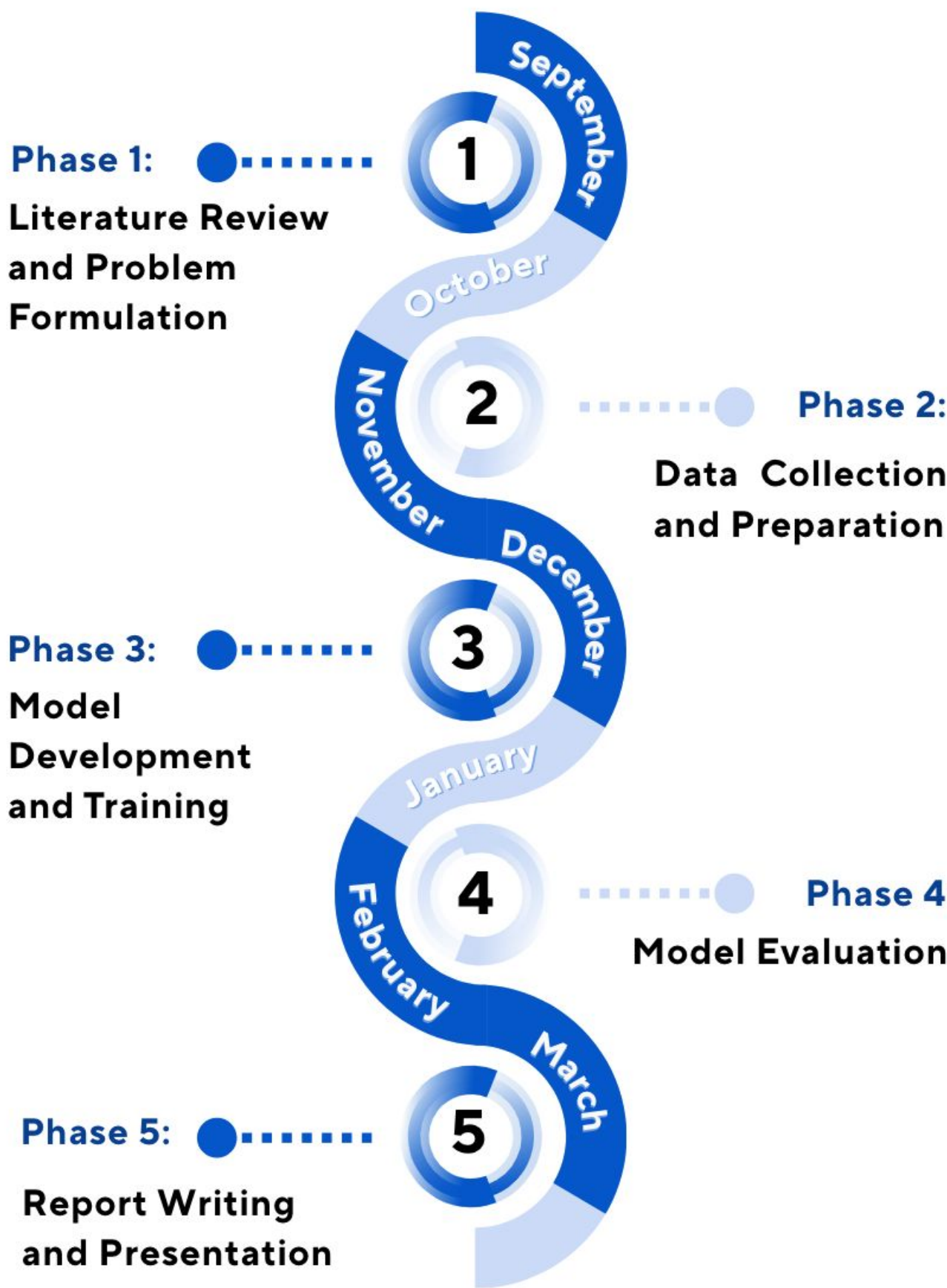
Experiment with quantization, batch sizes, and early stopping. Isolate variables to find optimal energy-efficient settings.

Compare energy, FLOPs, time, and F1 scores. Identify best efficiency-accuracy trade-offs across all models.

5. Expected Outcomes

This research will demonstrate that LoRA can reduce the power consumption of CNNs while preserving accuracy and robustness. By comparing standard, fine-tuned, LoRA-based, and lightweight models, we will reveal the trade-offs among training time, energy efficiency, and performance, and extend the study to adversarially trained models to show how LoRA alleviates computational costs. These findings matter because lowering training energy not only reduces environmental impact but also broadens access to advanced AI by making research and deployment more cost-effective, scalable, and equitable.

6. Timeline



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